

Sorting Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Implement every sort from scratch. Do NOT use built-in sort functions.

Section 1: Comparison-Based Sorts — Internals

★ **Bubble Sort — Implementation and Optimized Version (Early Exit)**

★ **Selection Sort — Implementation**

★ **Insertion Sort — Implementation**

★ **Merge Sort — Recursive Implementation**

★ **Quick Sort — Implementation with Lomuto Partition**

Quick Sort — Hoare Partition Scheme

Shell Sort — Implementation

★ **Heap Sort — Implementation using Max-Heap**

Section 2: Non-Comparison Sorts — Internals

★ **Counting Sort — Implementation**

★ **Radix Sort — LSD Implementation**

Radix Sort — MSD Implementation

★ **Bucket Sort — Implementation**

Pigeonhole Sort — Implementation

Section 3: Sorting Concepts and Stability Analysis

★ **Identify Stable vs Unstable Sorting Algorithms (Bubble, Merge = Stable; Quick, Heap, Selection = Unstable)**

★ **Identify In-Place vs Out-of-Place Sorts (Quick, Heap, Bubble = In-Place; Merge, Counting = Out-of-Place)**

★ **Time and Space Complexity Comparison of All Sorting Algorithms**

Demonstrate Stability: Sort Records by Secondary Key then Primary Key

When to Use Which Sort — Decision Framework

Section 4: Sorting Verification and Inversion Count

★ **Check if an Array is Sorted in Ascending Order**

Check if an Array is Sorted in Descending Order

★ **Count Number of Swaps in Bubble Sort**

★ **Count Inversions in an Array (using Merge Sort)**

Find Minimum Number of Swaps to Sort an Array

Section 5: Sorting Variants and Rearrangements

★ **Sort an Array of 0s, 1s, and 2s — Dutch National Flag**

★ **Sort by Absolute Difference from a Given Value**

★ **Sort Nearly Sorted (K-Sorted) Array using Min-Heap**

Sort Elements by Frequency (Most Frequent First)

Sort an Array based on Order of Another Array

★ **Wiggle Sort (Alternate High-Low Elements)**

Wave Array — Arrange in Wave Form

Section 6: Sorting Applied to Problems

★ **Merge Two Sorted Arrays without Extra Space**

★ **Find Median of Two Sorted Arrays**

★ **Meeting Rooms — Check if a Person can Attend All Meetings**

Meeting Rooms II — Minimum Number of Conference Rooms

★ **Largest Number Formed from Array of Numbers**

Minimum Difference between Largest and Smallest after K Modifications

★ **H-Index Calculation using Sorting**

Section 7: Merge Sort Patterns

★ **Count Inversions using Merge Sort**

External Sort Concept (Merge K Sorted Files)

★ **Reverse Pairs Count using Merge Sort**

Count of Range Sum using Merge Sort

Section 8: Quick Sort Patterns

★ **Quick Select — Kth Smallest / Largest Element**

Three-Way Quick Sort for Repeated Elements

Randomized Quick Sort

★ **Sort Colors (Dutch National Flag using Quick Sort Logic)**

Find the Kth Order Statistic

Section 9: Real-World Applications of Sorting

★ **Student Rank List Generator — Sort Students by Marks, then Alphabetically**

★ **E-Commerce Product Sorter — Sort by Price, Rating, or Relevance**

★ **Leaderboard Manager — Maintain Top-K Players using Heap Sort**

DNA Sequence Sorter — Sort Genetic Strings using Radix Sort

★ **Event Timeline Sorter — Sort Events by Start Time for Calendar View**

Log File Analyser — Sort Logs by Timestamp to Find Patterns

★ **Contacts Sorter — Sort Phone Contacts Alphabetically using Merge Sort**

Cricket Tournament Sorter — Rank Teams by Points, then NRR, then Name